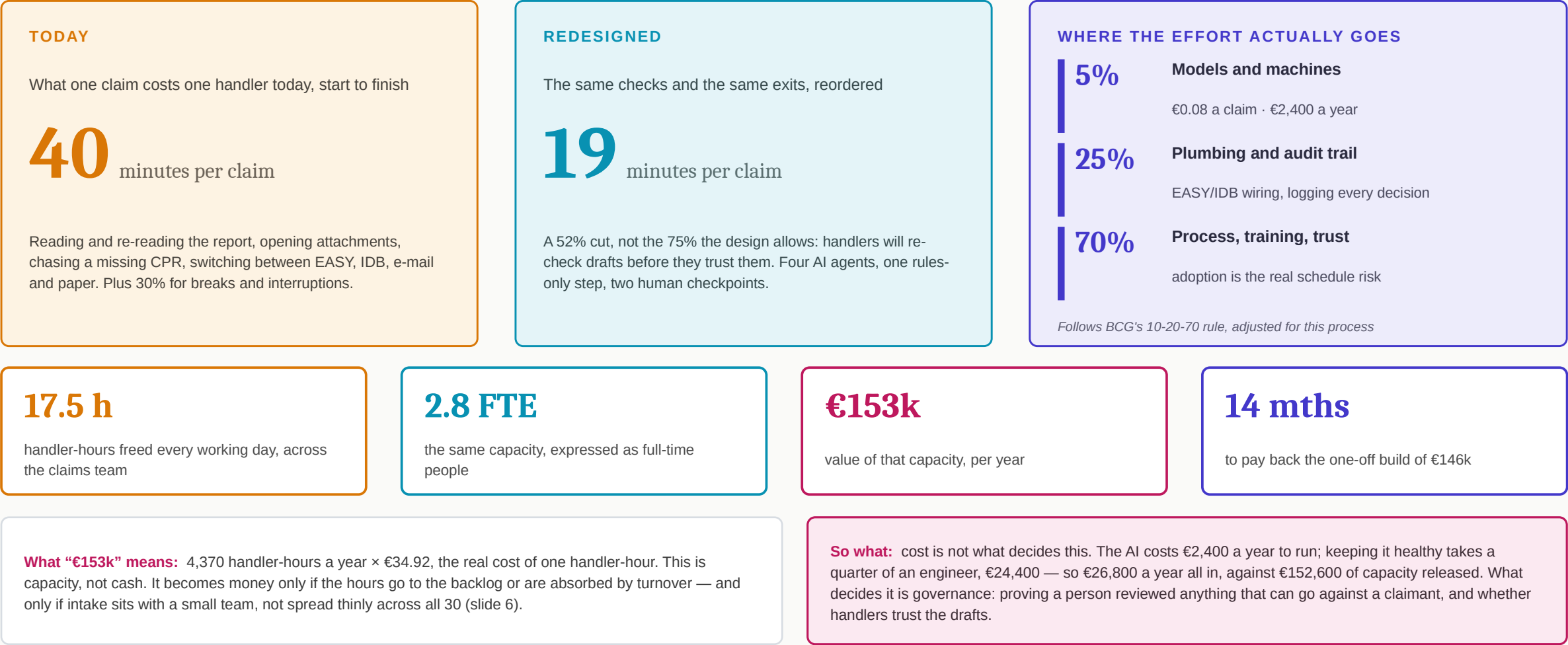


EXECUTIVE SUMMARY

Redesigning intake releases 17.5 handler-hours a day — the capacity of 2.8 full-time people

Two thirds of that work goes to claims that were always going to close quietly · a person still decides anything that can go against a claimant



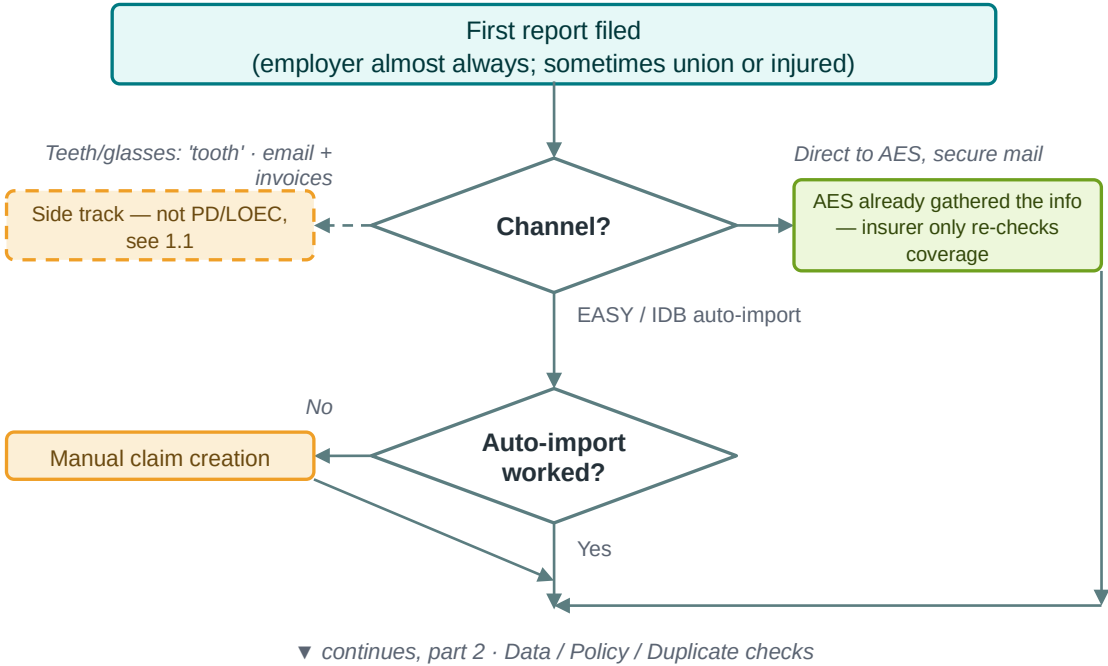
Conservative throughout: the 21-minute-per-claim saving is a three-point estimate (best 30 / likely 21 / worst 12), not the design maximum. Build-up on slide 6, sources on slide 7.

Three ways in and a separate side-track diagram become one classified intake

FNOL as it runs today (left) against the redesign (right) · first report to an opened, policy-linked claim

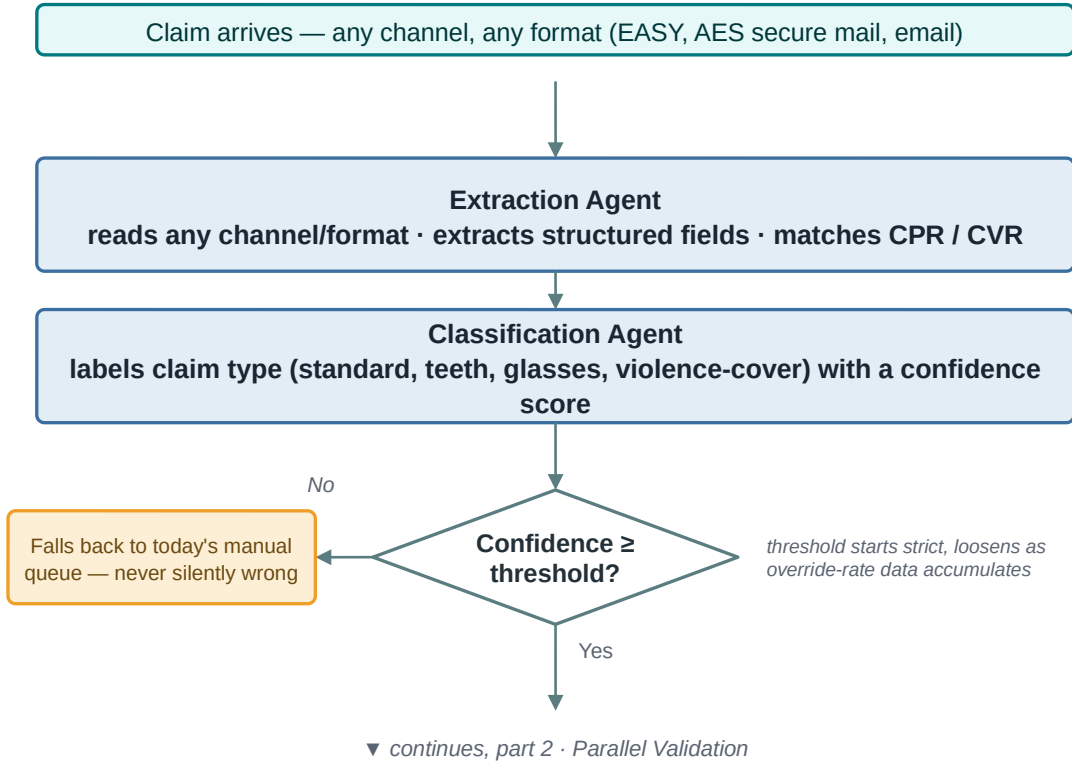
Trigger / start Step / decision Exception / rework Exit

TODAY



AI agent (LLM call) Rules only, no model Human checkpoint

AI-NATIVE — one LangGraph state graph · shared claim state carries the audit trail



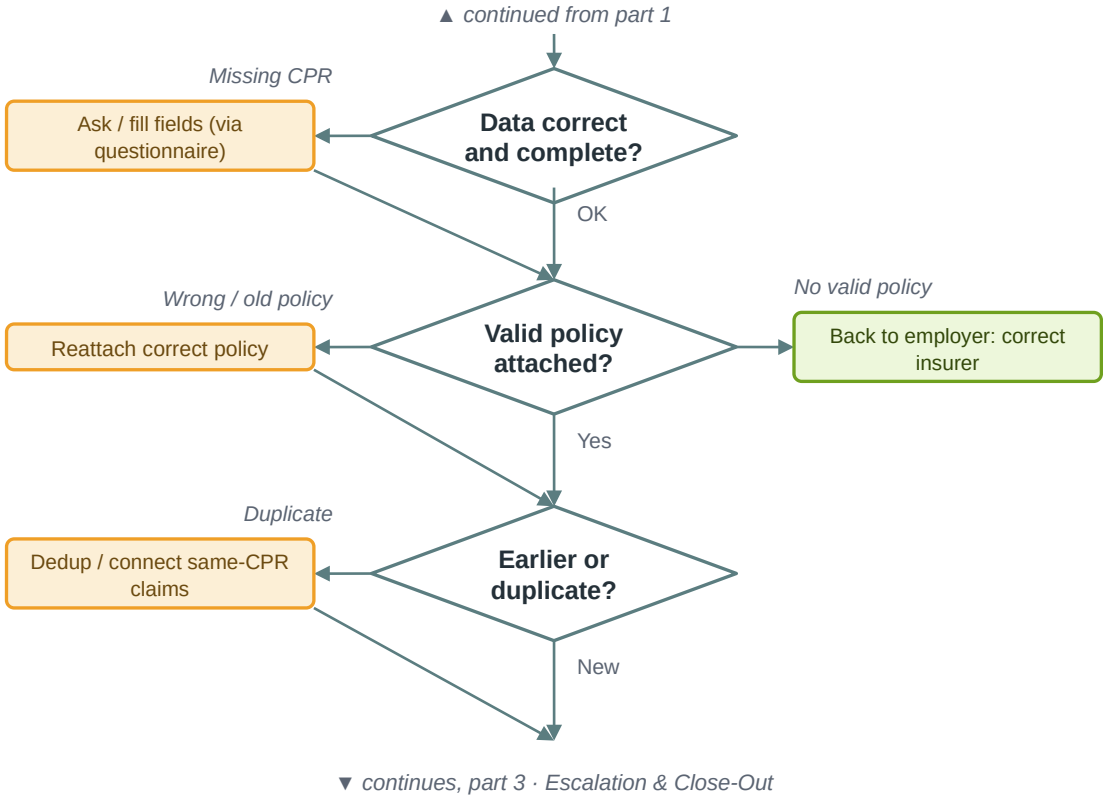
These two agents together replace 3 entry channels, the auto-import failure path, and the entire 1.1 side-track diagram.

Three checks done one after another become one check done all at once

Is it complete, linked to the right policy, and not already in the system? None of the three depends on the others

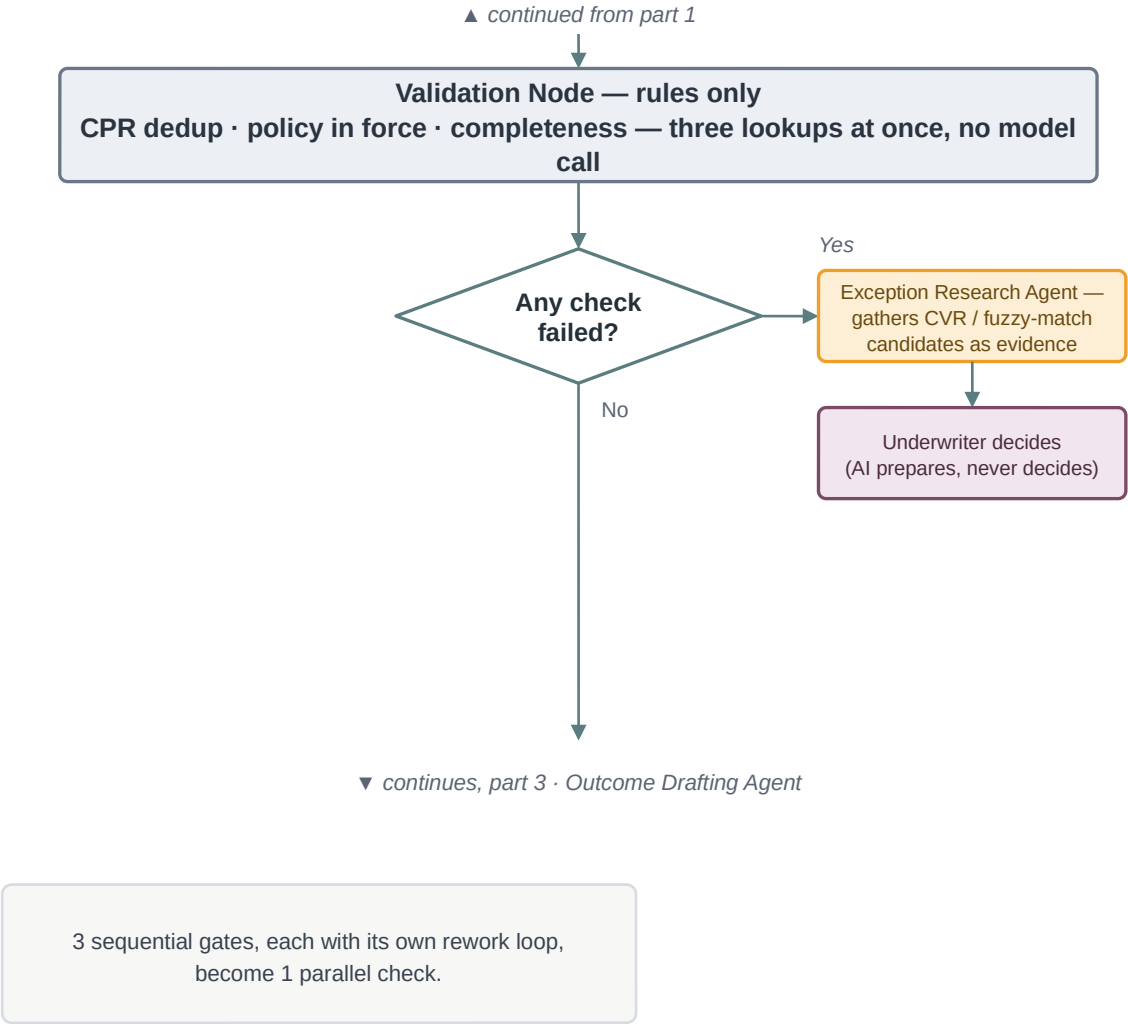
- Trigger / start
- Step / decision
- Exception / rework
- Exit

TODAY



- AI agent (LLM call)
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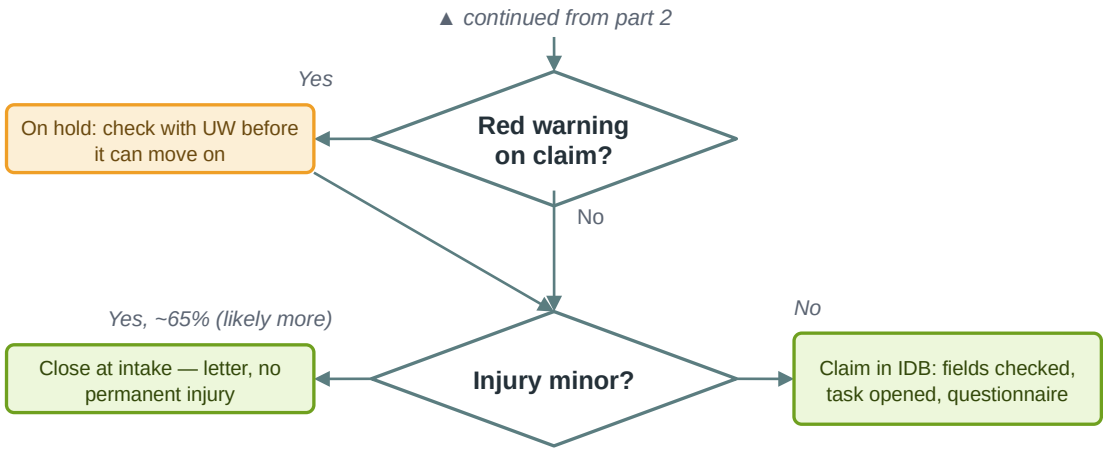


The AI writes the draft; the handler still makes the decision

From a system warning to a closed claim or a fully opened one — nothing that can go against a claimant is automated

- Trigger / start
- Step / decision
- Exception / rework
- Exit

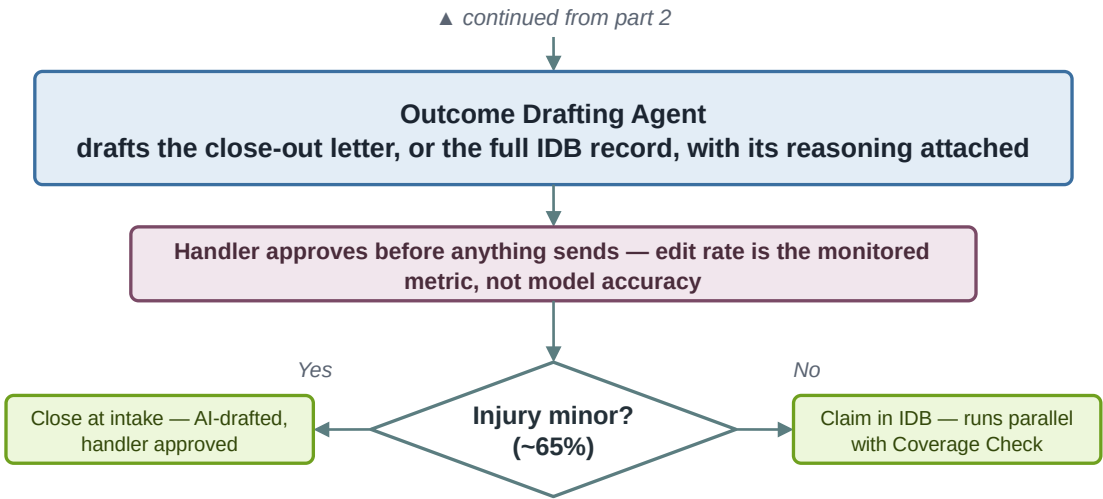
TODAY



Today: 7 decision gates, 3 entry channels, 10 exception or exit branches across the full FNOL flow.

- AI agent (LLM call)
- Rules only, no model
- Human checkpoint

AI-NATIVE — one LangGraph state graph · shared claim state carries the audit trail



Net result across all three parts:

4 LLM agents + 1 rules-only node · 2 human checkpoints kept by design · 3 entry channels → 1 · 3 sequential gates → 1 parallel check · shared claim state carries the audit trail

Coverage Check stops being a second investigation — the evidence is already in the claim

Every reason a claim can fail the coverage test, and where each one should be settled

Why coverage looks missing	Today	Redesigned	Who decides
Date sits inside a valid policy — no warning	quick manual check	rules, no model call	system
A valid follow-up policy exists	find it, reattach	registry lookup, reattach drafted	handler
Self-employed, voluntary cover	manual self-check	registry lookup, rules	handler
Group or branch policy not surfaced	ask underwriting	agent gathers the evidence, drafts the question	underwriting
Company-name mismatch	ask underwriting	agent fuzzy-matches CVR, drafts the question	underwriting
Wrong, expired, or no policy at all	back to employer	letter drafted, never sent unread	handler

Only two rows change hands. Four of the six are lookups against a register the insurer already queries — those become rules with no model call at all. The two that genuinely need judgement still go to underwriting, because that authority is not ours to take. The AI's job there is to arrive with the evidence assembled, not to answer the question.

WHAT THIS REMOVES

The second look at the policy	FNOL already asked “is a valid policy attached?”. Coverage Check asks it again, from the same documents.
The wait between the two	Today the claim sits in IDB until someone picks up the coverage step. It now runs beside claim creation, not after it.
Re-reading the file to ask underwriting	The question arrives with the policy history, the CVR match and the dates already attached.

Causes and their handling are taken from Diagram 2 of the brief · “who decides” is the design choice, not an observation

THE HANDOFF

The brief asks about the join between the two workflows. Today it is a queue. In the redesign there is nothing to hand over.

TODAY

- FNOL closes. The claim is written to IDB.
- It waits in a queue for the coverage step.
- A second person opens the same documents.
- They re-ask a question FNOL already asked.

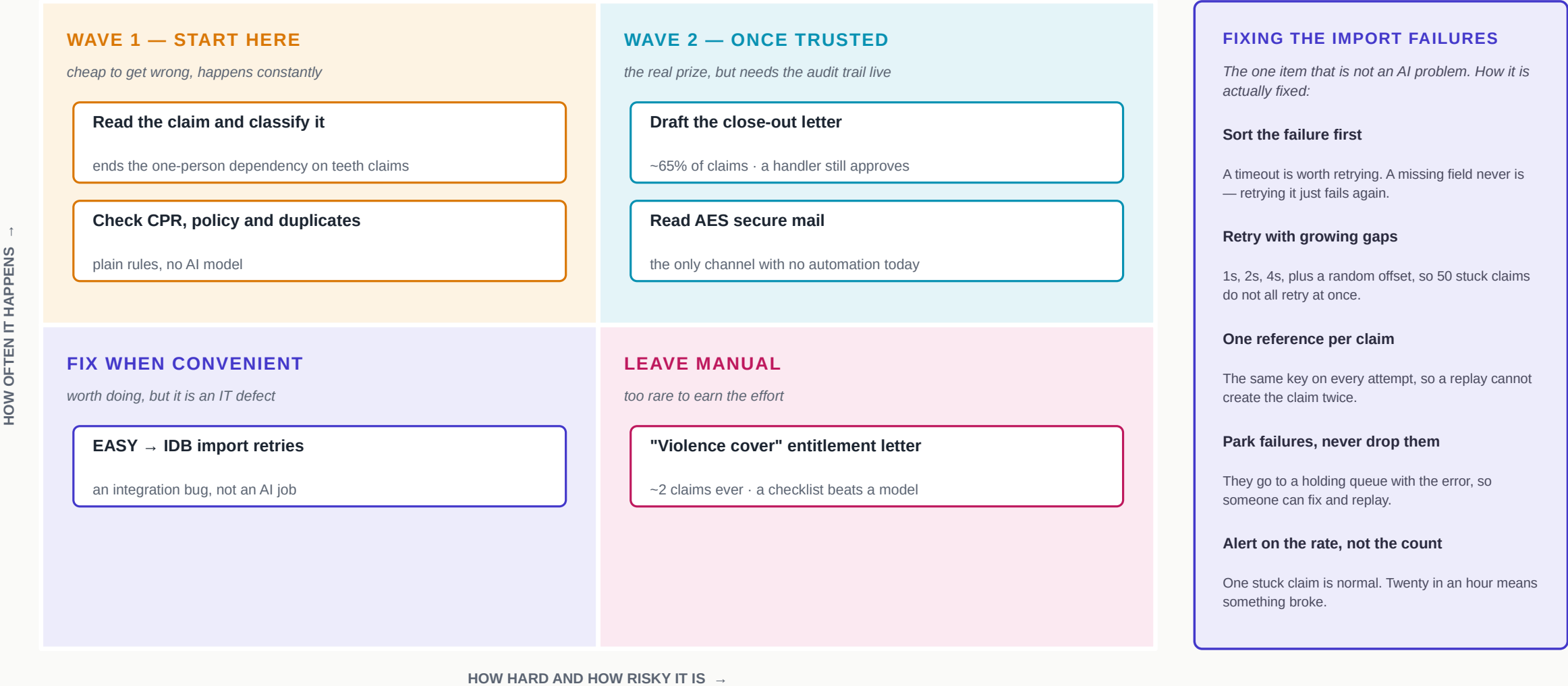
REDESIGNED

- One state graph spans both workflows.
- Extraction captures policy, dates, CVR and employment status once.
- Coverage is a rules pass over state that already exists.
- It runs beside claim creation, not after it.
- Only the exceptions become anyone's work.

A handoff you delete cannot be the thing that breaks.

Start where a mistake is cheap — that is what earns the trust the bigger wins depend on

“Hard” means build effort and regulatory exposure together, not just weeks of work



AES rulings and underwriting authority are deliberately absent — those decisions are not ours to make · frequencies from the case study

1 · SHADOW

wks 1–8

Agents run on live claims. Output is stored and compared with what the handler did. Nobody sees it.

Exit when extraction agrees with the handler on 95% of fields.

2 · SUGGEST

wks 9–16

The handler sees the draft and must edit or approve it. Nothing sends itself.

Exit when the edit rate on minor claims settles below 20%.

3 · APPROVE

mths 5–6

Minor claims arrive pre-filled. One click accepts. Everything else still drafted only.

Exit criteria reviewed monthly with the claims lead.

NEVER

—

No automatic decision on anything adverse to a claimant, on an AES ruling, or on underwriting authority.

This is a design boundary, not a phase.

WHAT WE WATCH — AND WHY THAT METRIC	
Edit rate on drafts	The trust signal. A model can be accurate and still not be used.
Field-level extraction precision	Checked against the source document, per field, not per claim.
Exception rate, by node	Says which step is actually struggling, not that “the AI” is.
Override rate by claim type	Feeds the confidence threshold. Teeth claims will differ from the rest.
First report to linked claim	The only number the business asked for. Everything else is diagnostic.
Alert on the rate, not the count — one stuck claim is normal, twenty in an hour is not.	

WHEN IT BREAKS

An agent fails, or confidence is low

The claim drops into today's manual queue. The old path is never removed, so the failure mode is a slower claim, never a wrong one.

A prompt or model version changes

200 historical claims are replayed first. A regression against the recorded outcome blocks the deploy.

One agent misbehaves in the wild

Each is behind its own switch. Turn off drafting and extraction keeps running.

The run dies half way

Graph state is checkpointed, so the claim resumes at the last good node instead of starting again.

Every decision, its inputs and its confidence are written to the audit log — that is what makes an adverse outcome defensible later.

Thresholds shown are the ones we would propose and then argue about with the claims lead — they are starting points, not findings

A · HANDLER TIME — MINUTES PER CLAIM, PER HANDLER

Step	Today	Redesigned
Read the report, re-read it, open attachments	10.0	0.7
Take notes, check fields, chase a missing CPR	6.1	2.8
Attach policy · duplicate check · coverage date	5.0	0.0
Decide, draft and send / open the full claim	5.5	3.0
Work an escalation or an underwriting hold	1.0	0.6
Switch between EASY, IDB, e-mail and paper	3.2	0.7
Touch time	30.8	7.8
Breaks, fatigue, interruptions, rework (+30%)	9.2	2.3
Standard time — what the design allows	40.0	10.1
What the business case actually banks	40.0	19.0

The design allows 10.1 min a claim. We do not bank that. The case uses a three-point estimate — (best 30 + 4 × likely 21 + worst 12) ÷ 6 = 21.0 min saved a claim — which is the 19.0 min above, and the 19 minutes on slide 1. In plain terms: we assume handlers capture 70% of the saving the design allows.

Cross-check: 40 min × 12,500 claims a year = 8,332 handler-hours — 17% of a 30-person team's yearly capacity. Plausible for the front end of a claims process. Had it come out at 60%, the model would be wrong.

D · THE BUSINESS CASE

EVERY YEAR

Capacity released	4,370 handler-h · 2.8 FTE
Worth, at €34.92 per handler-hour	€152,600 a year
Cost to run it — AI models and Azure	– €2,400
Ongoing support — 0.25 of an engineer	– €24,400
Net value, every year	€125,800

ONE-OFF

Build — 3 people × 6 months, fully loaded	€146,400
Payback	14 months

The build is loaded team capacity, not new cash — the AI team is already funded. New cash needed: €2,400 a year. A shorter build costs less: 4 months ≈ €98k.

B · AI COST TO PROCESS ONE CLAIM

Agent	Tokens in	Out	Runs	US\$ / claim
Extraction	11,500	700	100%	0.0314
Classification	900	120	100%	0.0032
Validation — rules only, no model	—	—	100%	0.0000
Exception research	2,500	400	20%	0.0020
Outcome drafting	2,200	550	100%	0.0110
Subtotal				0.0476
× 1.75 for retries, guardrails, evals				0.0833
Per claim				≈ €0.077

One claim ≈ a scanned form, an e-mail and 3–4 attachments. All four agents run on one model tier — gpt-5.6 terra, \$2 in / \$12 out per million tokens.

C · AZURE, WEST EUROPE — MONTHLY

Component	US\$/mth
4 container apps — 2 always on, 2 burst only	55
PostgreSQL B2s + 64 GiB storage and backup	61
Blob storage for documents + audit logs	8
Net new, incl. ~8% West Europe premium	134

Registry, networking, Key Vault and DevOps (≈ €670/yr) already exist, so they are not charged here. Azure DevOps gives repos and pipelines, not a database — PostgreSQL is genuinely new.

E · WHO ACTUALLY GETS THE TIME BACK

The brief says a support team does intake, not all 30 handlers — so the truth is in the middle rows. Our first question for GGW.

Intake handled by...	Claims each, per day	Intake work, per day	Time freed, per day
all 30 claims handlers	1.7	1.1 h	35 min
a support team of 12	4.2	2.8 h	88 min
a support team of 8	6.2	4.2 h	2.2 h
a support team of 6	8.3	5.6 h	2.9 h

Wage base Sachbearbeiter Schadenregulierung, €45,100 gross	StepStone Gehaltsreport
Employer on-cost ≈ €23 per €100 gross — health, pension, unemployment, care	Lohnnebenkosten 2026, sevdesk
Working time, breaks 38 h tariff week; statutory break rules	Arbeitszeitgesetz §4
Model prices gpt-5.6 terra — \$2 in / \$12 out per million tokens	OpenAI API pricing
Container hosting Consumption plan rates and the monthly free grant	Azure Container Apps pricing
Database PostgreSQL Flexible Server B2s — 2 vCore / 4 GiB	Azure Database for PostgreSQL
Build-team pay Senior ML engineer ≈ €98k; juniors at €70k as briefed	Glassdoor, ML Engineer Germany
Upkeep after go-live 15–20% of build effort a year — our 0.25 FTE is 17%	<i>Software maintenance rule of thumb</i>
Industry baseline 17 of 20 carriers still manual; none let AI decide alone	Decerto, 2026 Claims Decisioning Pulse
Where value comes from The 10-20-70 split — algorithms, technology, people	BCG, AI in customer service operations
Fixing import failures Retry vs. dead-letter, idempotency keys, alert on rate	Dead-letter queues and poison messages
Volumes and shares ~50 first reports a day · 65% minor · ~30 handlers	<i>GGW case study brief</i>

OUR ASSUMPTIONS — NOT SOURCED, AND THE FIRST THING WE WOULD TEST

Every per-step minute on slide 6 · the 30% allowance for breaks and interruptions · how often a CPR is missing (30%) or underwriting is asked (25%) · how often the redesign raises an exception (20%) · the tokens a typical claim needs · the €146k build · the 0.25 FTE upkeep · and how many people staff intake, which sets the per-person saving.

Two weeks of watching live intake would replace all of these with real measurements. That is step one of the plan, not an afterthought.

Prices checked September 2026 · US\$ converted at €0.92 · every euro figure is fully loaded — gross pay plus employer contributions